

SPIN-Seq: predicting the typed residue interaction network from sequence

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Abstract. *Knowing which residues are close does not reveal how they interact. Residue interaction networks (RINs) label edges by chemistry, but established generators read 3D structures. SPIN-Seq predicts the typed intra-chain RIN over eight interaction types from sequence alone, with Arpeggio labels used only as targets. Over 5,176,314 test pairs of a cluster-disjoint split, a frozen ESM-2 encoder and a 939k-parameter head reach macro AUPRC 0.520, against 0.110 for amino-acid propensity. A perfect contact map reaches only 0.132 and cannot rank types among contacts: its per-class AUPRC is prevalence over contact density, a constant 25-fold lift. The recipe was selected on this same evaluation, and the encoder was pretrained on a corpus likely containing the test sequences.*

1. Introduction

Recent structure predictors [Jumper et al. 2021, Lin et al. 2023] have greatly improved geometric modelling, but residue proximity does not specify interaction chemistry. Residue interaction networks (RINs) encode this chemistry, distinguishing salt bridges, aromatic stacking interactions and van der Waals contacts. Typed edges improve downstream models [Liu et al. 2026], yet established typed-RIN generators require 3D coordinates.

The immediate alternative is to fold the sequence and then profile the predicted structure, which may propagate errors where chemical typing is most sensitive. Many types depend on side-chain geometry, where dihedral error compounds outward from the backbone, from $\approx 14\%$ at χ_1 to $\approx 48\%$ at χ_3 and largest for polar residues [Maisuradze et al. 2025]: a backbone root-mean-square deviation near 1 Å does not guarantee a correct salt bridge. The central question is therefore how much interaction chemistry is recoverable from the sequence itself, under the constraint that every model input be sequence-derived and Arpeggio-derived labels serve only as targets.

Several related lines of research address parts of this problem, but not the full setting. Generators such as Arpeggio [Jubb et al. 2017] and RING [Del Conte et al. 2024] annotate typed interactions but read coordinates. Contact prediction returns geometry, not chemical type; the attention-derived map [Rao et al. 2021] is used here as an input channel, not as a competing method. Sequence-only methods predict *individual* types, including disulfide connectivity [Ceroni et al. 2006] and hydrogen bonds [Zhou et al. 2022], but

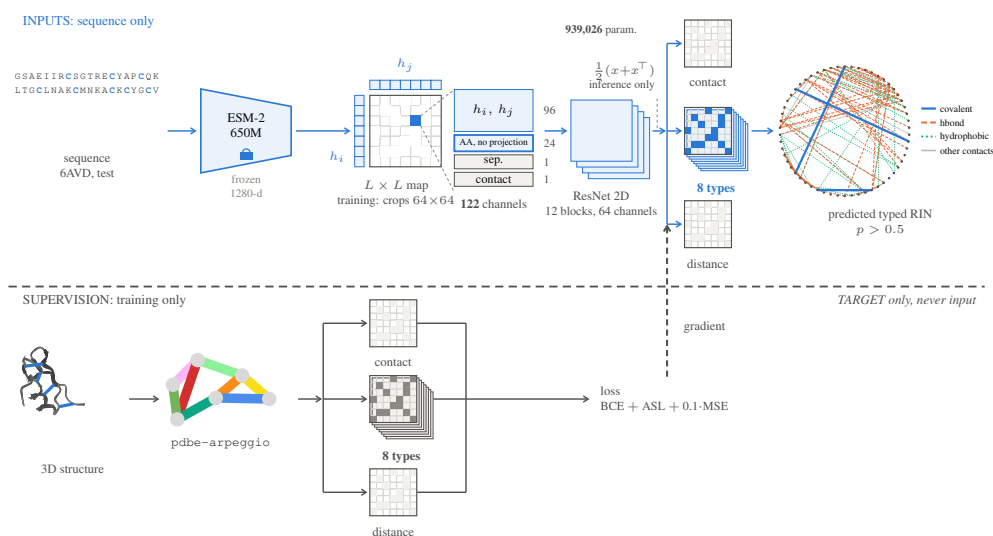


Figure 1. Above the barrier, the input path; below it, the supervision that pdbe-arpeggio derives from the 3D structure. At inference the lower half is absent. The three icons come from the same test chain (6AVD_A, 40 residues, 8 cysteines): the cysteines highlighted in the sequence are the nodes of the four disulfide bridges that reappear in the cartoon and as predicted covalent edges. The RIN on the right is model output ($p > 0.5$), not annotator label.

each resolves one class over a pair subset already restricted by a chemical rule. Sequence-based typing was shown for protein–ligand pairs [Pham et al. 2025], a setting that requires the ligand as input, and annotating a predicted structure is not direct prediction either. To the best of our knowledge, no *learned* method addresses the joint, multi-label prediction of intra-chain RIN types over the full graph, from one sequence, without alignment, templates or an intermediate structure.

The contributions are: **(i)** a formulation of sequence-only RIN prediction as multi-label classification over eight types, evaluated on all pairs under a cluster-disjoint split; **(ii)** evidence that the RIN reduces to neither composition nor a contact map, not even a perfect one; **(iii)** ablations and a learning curve that trace performance gains and point to data volume as the likeliest bottleneck.

2. Materials and Methods

2.1. Data and anti-leakage protocol

X-ray structures were retrieved from the Protein Data Bank (PDB) [Berman et al. 2000] using its search interface and filtered to resolution $\leq 2.5 \text{ \AA}$, one protein entity, 30–350 residues and one deposited model. Intra-chain scope is enforced at labelling time: one chain per entry, and an interaction is kept only when both atoms share the same `auth_asym_id` and Arpeggio reports `INTRA_SELECTION`. The most consequential decision is deduplication. One representative is retained from each 30%-identity cluster, so no retained chains share a cluster; the representatives are split into 3,962 training, 495 validation and 496 test chains, of which 494 are evaluated after two label-length mismatches are removed. Without it, near-identical homologues would fall on both sides of the split and inflate every metric.

Labels come from `pdbe-arpeggio` [Jubb et al. 2017] over mmCIF files: atoms map to residues and type labels are combined by logical OR for each pair. Water is discarded, as are all pairs with $|i - j| < 3$. Of the 11 raw types, eight are retained ($T = 8$): `xbond` and `metal` have zero examples, whereas `proximal` occurs on 99.995% of edges and is effectively redundant with contact existence. The difficulty follows from two scales that multiply. Only 4.01% of the 40,088,397 valid training pairs are edges, and within them the classes span a 392-fold range, from `polar` at 638,361 positives to `covalent` at 1,628. Classes are not mutually exclusive: the class counts sum to 105.19% of the edge count, or 1.05 types per edge. Training-set prevalences appear in Table 1.

2.2. Sequence-only inputs and architecture

Figure 1 separates the two paths that meet only at the loss. The sequence-derived inputs are the ESM-2 embedding, its attention-derived contact map, the amino-acid (AA) identities of the residue pair and $\log(1 + |i - j|)$; Arpeggio-derived annotations serve only as targets, entering solely through the supervised loss. Chains are encoded with ESM-2 650M [Lin et al. 2023] (layer 33, $d = 1280$), frozen, since the question is how much an off-the-shelf representation already encodes. The `PairConv2D` head projects 1280 to 32 dimensions, forms an $L \times L$ map by outer product and applies a dilated 2D ResNet, with heads for contact, for the $T = 8$ types and for distance. All the inputs but the embedding enter as extra channels of that map, *outside* the projection. The head has 939,026 trainable parameters, under 0.15% of the encoder.

Training uses 64×64 crops biased towards positives and rare classes, with per-class weights capped at 8 and the Asymmetric Loss (ASL) [Ridnik et al. 2021], which decouples the positive and negative focusing exponents and so suppresses abundant negatives without suppressing rare positives. The loss combines contact binary cross-entropy, type ASL and 0.1 times the distance mean squared error.

2.3. Evaluation protocol and baselines

Evaluation includes every valid test pair, without negative sampling: 5,176,314 pairs with 206,700 contacts (3.99%). The metric is macro AUPRC, the unweighted mean over the eight classes, so rare and frequent classes weigh equally. For AUPRC a random classifier scores the class prevalence, not 0.5, giving macro chance 0.005; 95% confidence intervals (CIs) come from bootstrapping *chains*, not pairs ($B = 1000$). One caveat governs the reading: the ablation ladder was scored under this same evaluation, so the final value is *selected on the test set*, with validation used only to monitor the train-validation gap. With no direct competitor, two baselines test the strongest alternative explanations. The contact oracle scores every type with the *true* contact label and cannot rank types among contacts, because its per-class AUPRC equals test-set prevalence divided by contact density, a *constant* $25.0 \times$ lift in all eight classes; it is a rescaled chance floor, and a ceiling only for binary-scoring methods. The propensity baseline estimates $P(t \mid aa_i, aa_j, \text{range}(|i - j|))$ from training counts. The contact-head score (P-CONTACT) is an ablation, not a baseline.

3. Results

3.1. Comparison with the contact map: information content and oracle

The information-theoretic result uses only the labels. Of the 206,700 test contact edges, 36.9% carry none of the eight types, and each edge carries 1.06 types on average. Summing the eight binary entropies conditioned on contact gives 3.209 bits per edge; because

Table 1. Per-class AUPRC on the 494 test chains. *Train prev.* is the training-set prevalence, given for the imbalance scale; the chance floor here is the test prevalence (Section 2.3). *Oracle*, *P-CONTACT* and *Prop.* are the perfect contact map, the contact head and the propensity table. **Blue is SPIN-Seq; **grey** is the strongest alternative in the row, the denominator of *Gain*.**

Class	Train prev.	Oracle	P-CONTACT	Prop.	SPIN-Seq [95% CI]	Gain
covalent	0.00004	0.001	0.001	0.092	0.801 [0.727, 0.868]	8.7×
aromatic	0.00041	0.009	0.006	0.077	0.461 [0.432, 0.490]	6.0×
hbond	0.00047	0.010	0.014	0.014	0.069 [0.049, 0.097]	4.9×
ionic	0.00056	0.014	0.011	0.083	0.394 [0.370, 0.416]	4.7×
carbonyl	0.00145	0.035	0.057	0.009	0.359 [0.339, 0.378]	6.3×
vdw	0.01166	0.292	0.425	0.210	0.681 [0.667, 0.695]	1.6×
hydrophobic	0.01172	0.295	0.212	0.098	0.599 [0.583, 0.615]	2.0×
polar	0.01592	0.401	0.669	0.299	0.796 [0.782, 0.809]	1.2×
macro	0.00528	0.132	0.174	0.110	0.520 [0.506, 0.535]	3.0×

label correlation is ignored, that sum is only an upper bound. The joint entropy is bounded below by the largest marginal, *polar* at 0.97 bit. A *perfect* contact map therefore leaves between 0.97 and 3.209 bits per edge unresolved, and both bounds are far from the zero that would make the RIN a relabelling of proximity.

The sequence-only model reaches macro AUPRC 0.520 (95% CI [0.506, 0.535]) against 0.132 for the oracle, a factor of 3.9. However, the macro average hides the class-wise pattern (Table 1). Except for *hbond*, the model-to-oracle ratio grows with class rarity: 2.0× in *polar*, 28× in *ionic*, 51× in *aromatic* and 800× in *covalent*. This model-to-oracle gap is not an artefact of the binary score: the continuous P-CONTACT performs better (0.174) but remains 3.0× below. Against the *strongest* alternative in each class the model outperforms all eight, by margins from 1.2× to 8.7×.

3.2. Comparison with amino-acid propensity

A second alternative is compositional: since *ionic* bonds pair an acidic residue (D/E) with a basic one (K/R/H), and *covalent* bonds join two cysteines, a lookup table might reproduce the result without context. The model outperforms the propensity baseline by +0.410 (0.520 against 0.110, 95% CI [0.395, 0.424]); all nine comparisons are significant at the resampling floor $p = 1/B = 0.001$. That baseline is strong exactly where chemistry predicts, beating the oracle on *ionic* (0.083), *aromatic* (0.077) and *covalent* (0.092): the explanations are complementary and the model beats both. Treating the chemical rules as filters clarifies why. Cys–Cys pairs cover 97.7% of the *covalent* edges but only 7.7% of candidate Cys–Cys pairs bond; the D/E–K/R/H rule covers 99.5% of the *ionic* edges, yet only 1.5% of its 189,006 admissible pairs are salt bridges. Composition says fairly precisely *where to look* and little about *which pair to select*. Combining the two was not evaluated, and may account for part of the gain.

3.3. Ablation, model capacity and learning curve

A pairwise multilayer perceptron (MLP) on the same frozen embedding already reaches macro 0.429, itself 3.25× the perfect-contact oracle, and the remaining steps add 0.091: the thesis does not rest on the head architecture. Within the ladder the two tasks behave as largely separable (Table 2). Step 1 accounts for almost the entire contact gain and almost none of the chemistry; the later steps leave contact flat at 0.815 while the macro climbs

Table 2. Ordered ablation ladder on 494 test chains. *Contact* and *Macro* are AUPRC; the ladder is descriptive, since the final recipe was selected on this same evaluation. Blue is the headline metric, red the only decreases, green their recovery once the explicit AA-pair feature is added.

Step	Change	Contact	Macro	ionic	aromatic
0	Pair MLP	0.718	0.429	0.363	0.363
1	$L \times L$ map + ResNet	0.814	0.440	0.248 ↓	0.120 ↓
2	Class-weight cap $3 \rightarrow 8$	0.815	0.460	0.270	0.204
3	ASL instead of focal loss	0.815	0.478	0.298	0.289
4	AA pair feature	0.815	0.520 [0.506, 0.535]	0.394 ↑	0.461 ↑

from 0.440 to 0.520. The 1280-to-32 projection of step 1 omits the explicit AA-pair feature, and *ionic* and *aromatic* fell at the step that produced the largest contact gain; these were the ladder’s only decreases. Adding that feature *outside* the projection (step 4) raised the macro from 0.478 to 0.520, concentrated in the classes the chemical hypothesis predicts (*aromatic* +0.172, *ionic* +0.096, *polar* within seed noise) rather than spread uniformly, as generic capacity would give.

Widening the projection to 128 reduced contact/macro AUPRC to 0.801/0.500 with no class improving, and the train–validation gap rose from 0.026 to 0.058. The learning curve over nested training subsets slows early but shows no resolved saturation: the slope with respect to $\ln(n)$ falls from 0.076 to 0.037, while the third segment, 0.038, is indistinguishable from the second within seed noise, with the gains from added data concentrated in the rare classes.

The selection query leaves 5,926 unused clusters; including them would yield $2.2\times$ as many clusters as the 4,953 used here, without relaxing the filters. A log-linear fit over four points ($R^2 = 0.967$) extrapolates to ≈ 0.57 and the last observed segment to ≈ 0.55 , but neither the functional form nor saturation is resolved. Over three seeds the recipe yields macro 0.518 ± 0.003 (standard deviation across seeds, not the bootstrap CI); the 0.520 quoted throughout is the reference seed, and differences below ≈ 0.01 are treated as unresolved.

4. Discussion

The value 0.520 must be read against a macro chance of 0.005 and a *perfect* contact map at 0.132. The *hbond* class is the main low-performing one, and its result (0.069) is treated here as an open challenge; excluding it, the mean over the other seven is 0.584. Re-deriving labels with Arpeggio’s hydrogen-minimisation protocol over 119 structures changes *hbond* only modestly (Jaccard 0.967, against exactly 1.000 for the other eight non-empty Arpeggio types, *proximal* included), so that perturbation does not explain the low AUPRC. The final data increase gave no resolved gain either: although *hbond* has $11.6\times$ more examples than *covalent*, its AUPRC is about one eleventh as large. Under this representation, architecture and annotation convention the class remains poorly resolved, and the evidence does not distinguish a limit of sequence information from limitations of the frozen readout or of Arpeggio’s convention.

Five limitations bound the results. The analysis covers 30–350-residue single chains resolved by X-ray at $\leq 2.5 \text{ \AA}$, and the labels come from one annotation tool, not from direct measurement. The frozen encoder was pretrained on UniRef, which likely

contains the test sequences, and its contact readout was fitted on PDB structures, so the cluster-disjoint split removes leakage into the trained head, not into the representation. The eight-type vocabulary is not exhaustive: 36.9% of test contacts receive none. Finally, the fold-and-profile route was not run, so that argument remains untested on the same chains.

5. Conclusion and future work

The results show that the typed RIN is predictable from sequence alone and is a target distinct from the contact map. On the reference seed and over 5,176,314 pairs, SPIN-Seq reaches macro AUPRC 0.520 against 0.132 for a perfect contact map, with the gap over the oracle generally widening as classes become rarer. Locating a contact and typing it behave as separable, and the largest chemical gain occurs when the explicit AA-pair feature is added outside the projection.

Since the tested width increase produced no gain and the learning-curve slope has not resolvably fallen to zero, the priority is to expand training with the 5,926 unused clusters while reserving a fresh cluster-disjoint test split, since the current test chains already informed recipe selection. Future work will also compare SPIN-Seq with the fold-and-profile route. Code, per-split chain lists and raw evaluation outputs are available at <https://github.com/labitec-cloud/SPIN-Seq>.

References

- Berman, H. M., Westbrook, J., Feng, Z., Gilliland, G., Bhat, T. N., Weissig, H., Shindyalov, I. N., and Bourne, P. E. (2000). The protein data bank. *Nucleic Acids Research*, 28(1):235–242.
- Ceroni, A., Passerini, A., Vullo, A., and Frasconi, P. (2006). DISULFIND: a disulfide bonding state and cysteine connectivity prediction server. *Nucleic Acids Research*, 34(suppl_2):W177–W181.
- Del Conte, A., Camagni, G., Clementel, D., Minervini, G., Monzon, A., Ferrari, C., Piovesan, D., and Tosatto, S. (2024). Ring 4.0: faster residue interaction networks with novel interaction types across over 35,000 different chemical structures. *Nucleic Acids Research*, 52:W306–W312.
- Jubb, H. C., Higuero, A. P., Ochoa-Montano, B., Pitt, W. R., Ascher, D. B., and Blundell, T. L. (2017). Arpeggio: A web server for calculating and visualising interatomic interactions in protein structures. *Journal of Molecular Biology*, 429(3):365–371.
- Jumper, J., Evans, R., Pritzel, A., et al. (2021). Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589.
- Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., Smetanin, N., Verkuil, R., Kabeli, O., Shmueli, Y., dos Santos Costa, A., Fazel-Zarandi, M., Sercu, T., Candido, S., and Rives, A. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130.
- Liu, L., Jiang, Y., Gu, Y., Zhao, S., and Ding, Y. (2026). Rit-hedge: A residue interaction type-aware heterogeneous graph-embedding model for predicting protein thermal stability. *Neural Networks*, 199:108707.
- Maisuradze, G. G., Thakur, A., Khatri, K., Haldane, A., and Levy, R. M. (2025). Predicting side chain conformations in folded proteins by alphafold: Perspective and challenges. *Biophysical Journal*, 124(21):3755–3771.
- Pham, P., Nguyen, V. T. D., and Hy, T.-S. (2025). Linker: Learning interactions between functional groups and residues with chemical knowledge-enhanced reasoning and explainability. *Journal of Chemical Information and Modeling*. Preprint: arXiv:2509.03425.
- Rao, R., Meier, J., Sercu, T., Ovchinnikov, S., and Rives, A. (2021). Transformer protein language models are unsupervised structure learners. In *International Conference on Learning Representations (ICLR)*.
- Ridnik, T., Ben-Baruch, E., Zamir, N., Noy, A., Friedman, I., Protter, M., and Zelnik-Manor, L. (2021). Asymmetric loss for multi-label classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 82–91.
- Zhou, S., Liu, Y., Wang, S., and Wang, L. (2022). Effective prediction of short hydrogen bonds in proteins via machine learning method. *Scientific Reports*, 12(1):469.